

Software Testing Class Pre-Session Assessment

A short, mandatory questionnaire for university students covering background and core software testing concepts.

* Indicates required question

1. *
Consent to Participate

Check all that apply.

☐ I voluntarily agree to participate in this study and understand that my responses are anonymous and will be used only for research purposes.

2. **Please create a unique participant code using the following format:** *
last 2 digits of phone number + birth month + first letter of mother's first name
(e.g., 45 + 07 + C → 4507C)

Use the same code in all questionnaires.

SECTION A: Background Information

3. 1. What is your academic field ? *

Mark only one oval.

- ☐ Computer Science / Software Engineering/ Computer Engineering
- ☐ Information Technology
- ☐ Other (please specify):

4. 2. How would you rate your programming experience? *

Mark only one oval.

- ☐ None
- ☐ Basic
- ☐ Intermediate
- ☐ Advanced

5. 3. How would you rate your current experience with software testing concepts and practices? *

Mark only one oval.

- ☐ None
- ☐ Basic
- ☐ Intermediate
- ☐ Advanced

6. 4. How would you rate your debugging experience? *

Mark only one oval.

- ☐ None
- ☐ Basic
- ☐ Intermediate
- ☐ Advanced

7.

*

5. Have you previously used software testing tools or frameworks (e.g., Selenium, JUnit, Cypress)?

Mark only one oval.

- ☐ No experience
- ☐ Yes, in coursework
- ☐ Yes, in personal/academic projects
- ☐ Yes, in professional/industry context

SECTION B: Core Knowledge Check (Pre-Class Assessment)

8. 6. A program is supposed to compute the sum of two numbers. For input (2, 3), it returns 6.

*

What are you doing when you notice this problem?

Mark only one oval.

- ☐ Debugging
- ☐ Testing
- ☐ Development
- ☐ Compilation

9. 7. After detecting the issue, you inspect the code and find: *

```
return a + b + 1;
```

What are you doing now?

Mark only one oval.

- ☐ Debugging
- ☐ Testing
- ☐ Integration testing
- ☐ Execution

10. 8. Which scenario represents unit testing? *

Mark only one oval.

- ☐ Testing the full login system
- ☐ Testing a single function in isolation from other components
- ☐ Testing system performance
- ☐ Testing user interface

11. 9. Which scenario represents integration testing? *

Mark only one oval.

- ☐ Testing a single function
- ☐ Testing UI design
- ☐ Testing interactions between components
- ☐ Testing execution speed

12. 10. A tester checks an ATM by inserting a card and entering a PIN, without knowing how the system is implemented. *

What type of testing is this?

Mark only one oval.

- ☐ White-box testing
- ☐ Debugging
- ☐ Unit testing
- ☐ Black-box testing

13. 11. Which activity represents white-box testing? *

Mark only one oval.

- ☐ Testing only inputs and outputs
- ☐ Designing tests based on internal code logic
- ☐ Testing without code access
- ☐ Testing UI behavior

14. 12. Which example represents functional testing? *

Mark only one oval.

- ☐ Checking system performance
- ☐ Testing security vulnerabilities
- ☐ Verifying that login works correctly
- ☐ Evaluating user interface design

15. 13. A testing technique intentionally introduces faults into the code to check whether existing tests can detect them. *

What concept does this represent?

Mark only one oval.

- ☐ Mutation testing
- ☐ Integration testing
- ☐ Regression testing
- ☐ Debugging

16. 14. Consider the following code: *

```
int sum(int a, int b) {  
    return a - b;  
}
```

Which valid test case is most effective at revealing the bug?

Mark only one oval.

- ☐ assertEquals(0, sum(0,0))
- ☐ assertEquals(2, sum(1,1))
- ☐ assertEquals(1, sum(3,2))

17. 15. A test case fails after a recent code change. What is the most appropriate first step? *

Mark only one oval.

- ☐ Delete the failing test
- ☐ Immediately rewrite the entire code
- ☐ Investigate whether the failure is due to a bug in the code or the test
- ☐ Ignore the failure if the program still runs

18. 16. If all tests pass but the system still contains a bug, what does it indicate? *

Mark only one oval.

- ☐ The system is correct
- ☐ The tests are not effective enough to detect the bug
- ☐ Debugging has already been completed
- ☐ The compiler is incorrect

SECTION C: Self-Confidence

19. 17. Please indicate your level of agreement with the following statements: *

Mark only one oval per row.

	1 (Strongly disagree)	2	3 (Neutral)	4	5 (Strongly agree)
I understand the basic concepts of software testing and debugging	☐	☐	☐	☐	☐
I can write a simple test case.	☐	☐	☐	☐	☐
I can identify and debug software errors.	☐	☐	☐	☐	☐
I feel confident analyzing test failures.	☐	☐	☐	☐	☐

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